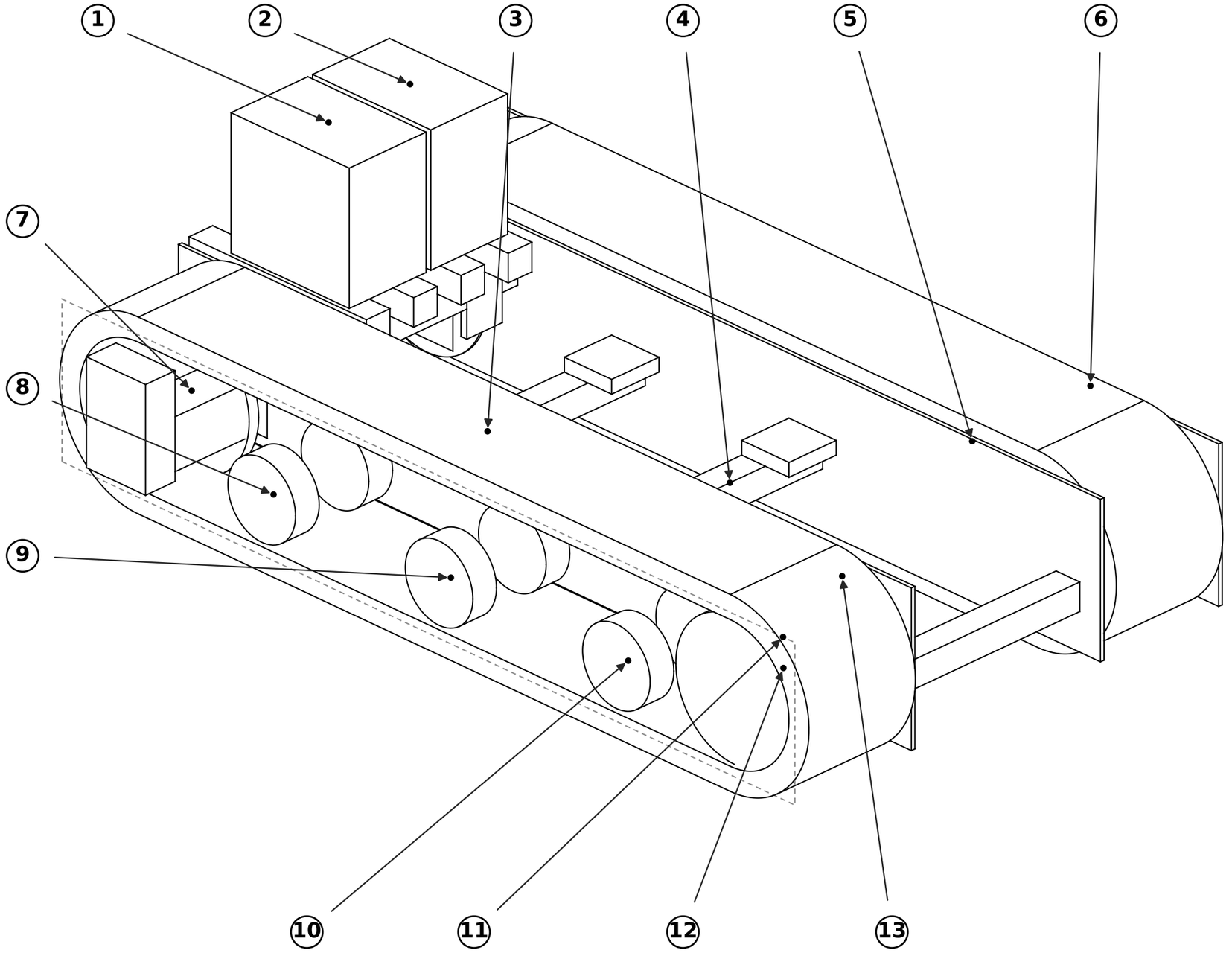
UC-V2 — ISOMETRIC VIEW (NTS, SIMPLIFIED GA)



Near track shown; the far track is behind the far inner side plate — see GA-005 for both tracks.
Visible edges only; dimensions mm; FRONT = +X (right).

LEGEND

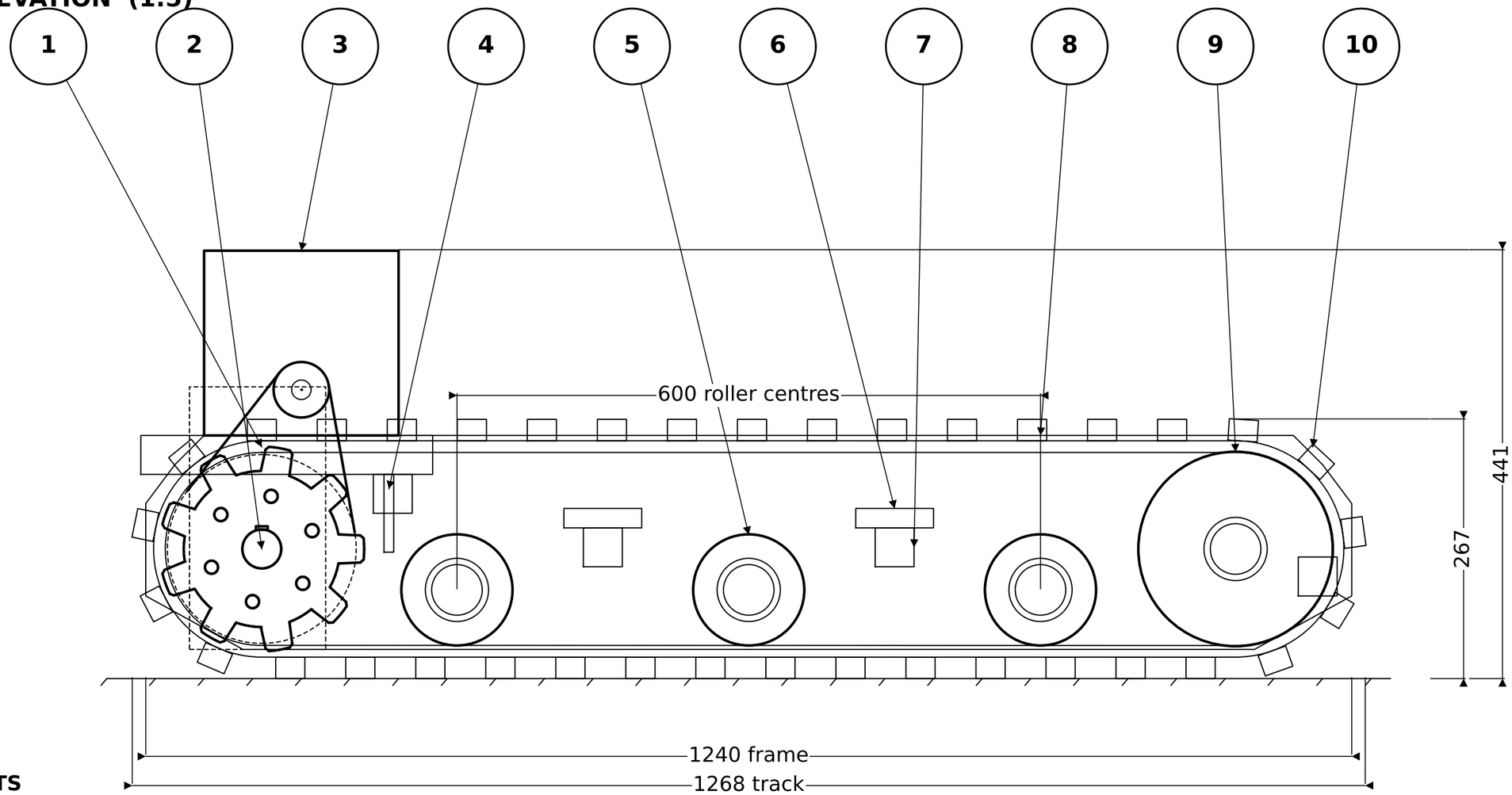
1. R MOTOR
2. L MOTOR
3. TRACK — 180 × 72 × 37
4. MAIN FRAME
5. L INNER SIDE PLATE (Y+)
6. L OUTER SIDE PLATE (Y+)
7. TRACK SPROCKET 72z9
8. ROLLER STATION 1
9. ROLLER STATION 2
10. ROLLER STATION 3
11. R OUTER SIDE PLATE (Y-) - dotted
12. IDLER STATION
13. R INNER SIDE PLATE (Y-)

L = +Y (far side); R = −Y (near side).

R outer side plate removed - dotted outline.

Simplified GA; not for manufacture.

SIDE ELEVATION (1:5)



CALL-OUTS

1	SPROCKET 72z9	**6**	DECK PAD 80 × 80
2	AXLE Ø40	**7**	BEAM SHS40
3	MOTOR 1.2 kW	**8**	SIDE PLATE 6 mm
4	LIFT LUG	**9**	IDLER Ø200
5	ROLLER Ø114.3 (3)	**10**	TRACK 180 × 72 × 37

NOTES

1. Near side plate removed for clarity; running gear shown true.
2. Datum: ground at lug tips z=0, X travel, Z up. Dimensions mm.
3. Track loop: 37 links × 72 pitch, R105.26, C1001.3 (closure 0).
4. Chain plane y=±140; 48T/14T 428, 60 links, C=168.66 mm.
5. Overall W826 over UCFL208 flanges; side-plate span 726; track span 700.
6. Motor body y 4..134 (inboard); chain runs at y=140, outboard of motor.
7. Loop centre z=133.3; roller centre z=91.2. Call-outs 1-10 at left.

FLAIL MOWER 500 / UC-V2

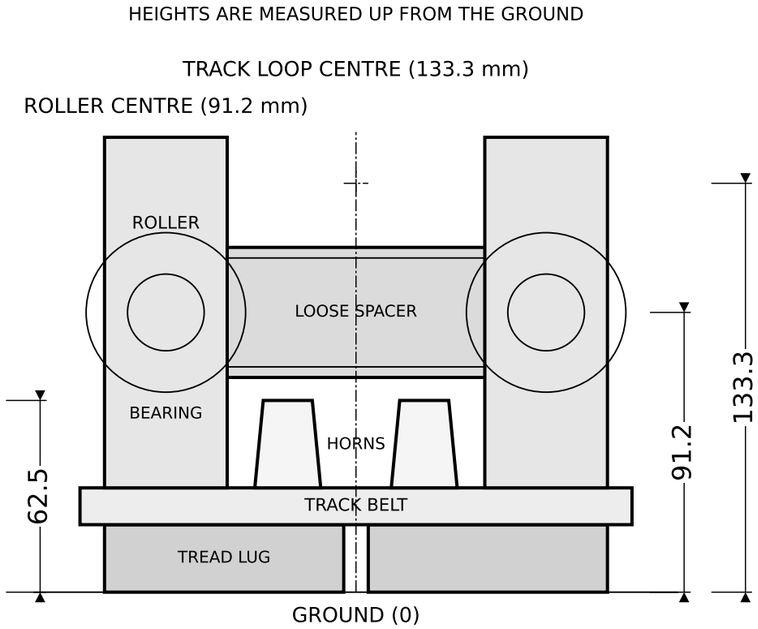
GENERAL ARRANGEMENT - SIDE ELEVATION

DWG No.
UCV2-GA-001
REV A

MATERIAL	QTY	UNITS
-	1	**mm**
		1st ANGLE

SCALE	SIZE	SHEET	DATE	DRAWN
1:5	A3	1 / 15	2026-10-29	OC

TRACK CROSS-SECTION & ROLLER HEIGHTS (SCHEMATIC, 1:2)



TERMS IN PLAIN LANGUAGE

term	meaning
Ground level	Bottom of the track lugs; all heights start here (0).
Side plate	6 mm vertical side panel; 4 off: inner and outer.
L / R	L = +Y (far side); R = −Y (near side).
Track loop centre	Centre of the oval track path; 133.3 mm above ground.
Roller centre	Centre of a roller drum; 91.2 mm above ground.
Track inner surface	Inside face of the rubber track; rollers run on it; 34 mm above ground.
Guide horns	Two raised ribs inside the track; 66 mm overall, 23 mm gap; top at 62.5 mm.
Drum spacing	Two drums per station, one at ±62 mm from track centre.
Track belt	Rubber band around the wheels; rollers run on its inner face.
Tread lug	Raised rubber block on the outside that grips the ground.
Roller drum	Cylindrical roller that supports the track; two per station.
Drum backplate	6 mm disc welded inside the inner end of each drum.
Bearing	6205-2RS (25 × 52 × 15 mm), one per drum; runs on Ø25 shaft.
Bearing seat	Welded ring that holds the bearing; 65/52 × 22 mm.
Press fit	Bearing is pressed in; no adhesive.
Loose spacer	Tube between the two bearings; 42.4/35.5 × 109 mm.
Open end	Outer end of the drum is open; plastic cap optional.
Unless noted	All dimensions in mm; 1st-angle projection.

NOTES

1. All dimensions in millimetres; 1st-angle projection.
2. Ground is the bottom of the track lugs; heights go up from there.
3. The guide horns are the two raised centre ribs inside the track.
4. Bearing seat and shaft are precision-fitted.
5. Bearings are press-fitted; no adhesive.

FLAIL MOWER 500 / UC-V2

DEFINITIONS & STANDARD NOTES

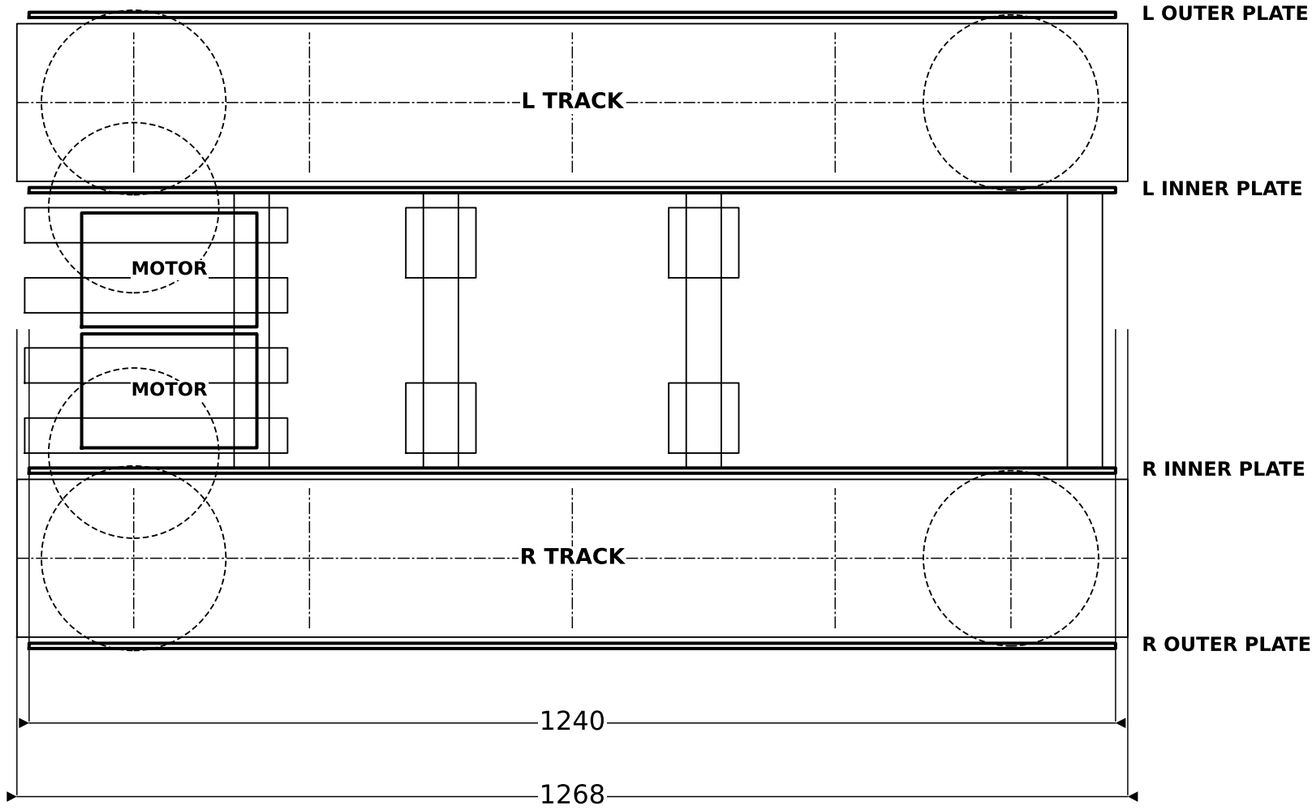
DWG No.
UCV2-P-011
REV A

MATERIAL: -
QTY: -

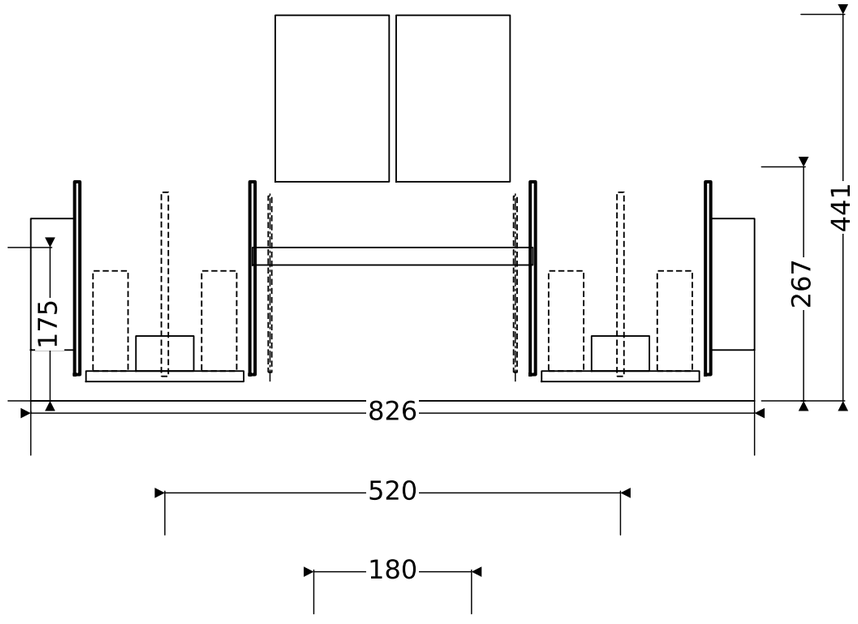
UNITS: mm
1st ANGLE

SCALE: 1:2
SIZE: A3
SHEET: 15 / 15
DATE: 2026-10-29
DRAWN: OC

PLAN (1:7)



END VIEW (1:7)



NOTES

1. Track centres 520; side-plate gauge 200; overall 826 over UCFL.
2. Deck pads: x=±150, y=±100 mm; top z=175 mm (4× M12).
3. Motors: 2× 1.2 kW on rails z250..441; pinions y=±140.
4. Lift lugs: x=-370, y=±100.
5. Tracks y=±260, 180 wide; inner side plates y=±160; outer y=±360.
6. L=+Y (far side), R=-Y (near side); four side-plate lines in end view.

FLAIL MOWER 500 / UC-V2

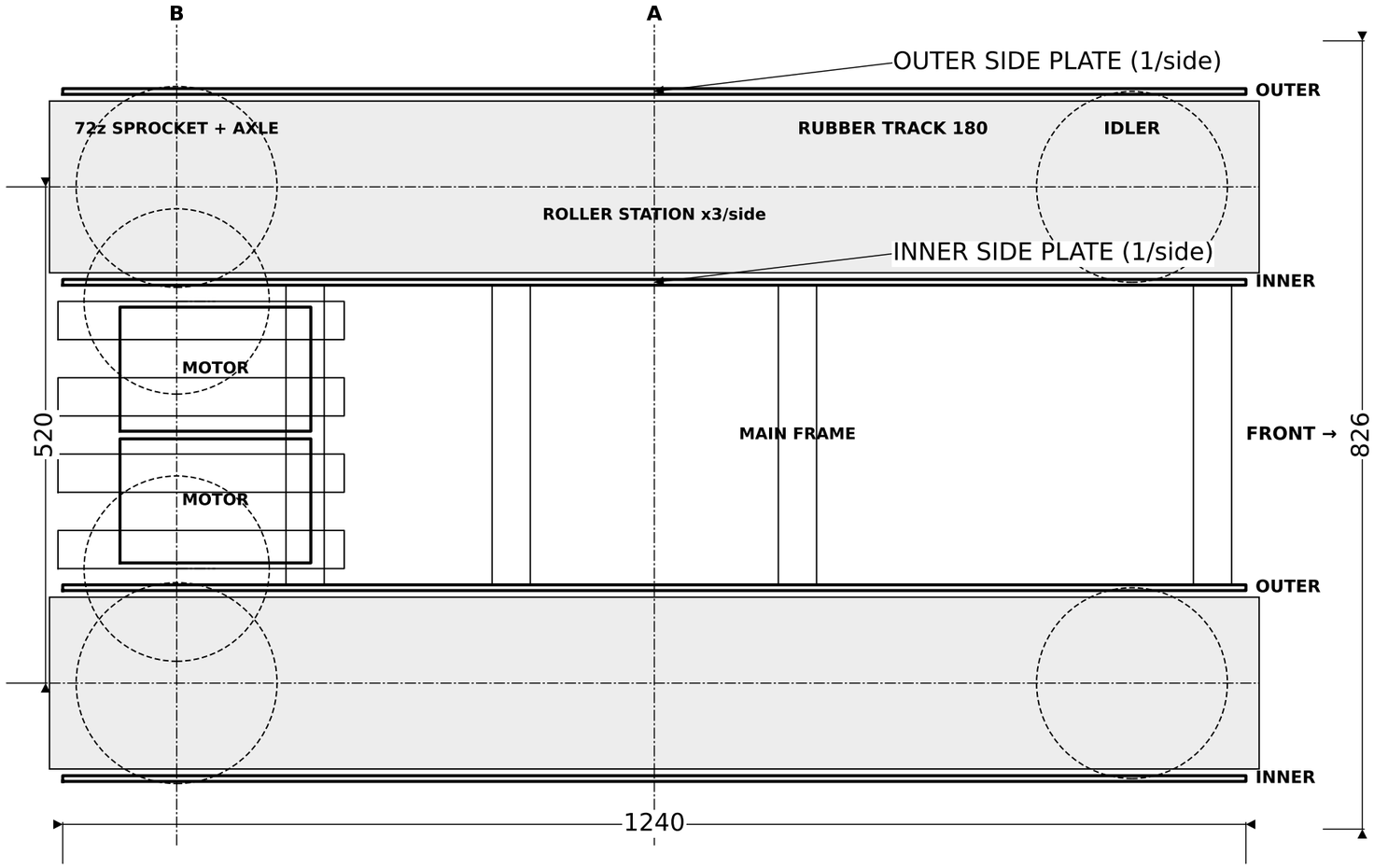
GENERAL ARRANGEMENT - PLAN & END VIEW

DWG No. **UCV2-GA-002** REV A

MATERIAL	QTY	UNITS
-	1	**mm** (1st ANGLE)

SCALE	SIZE	SHEET	DATE	DRAWN
1:7	A3	2 / 15	2026-10-29	OC

TOP VIEW (1:6)



NOTES

1. Top view; both side plates per side. L=+Y (top), R=-Y (bottom).
2. Inner side plates at y=±160; outer at y=±360 (6 mm).
3. Track centres 520; side-plate gauge 200; overall 826 over UCFL.
4. Section A-A at x=0 (roller station); B-B at x=-500.6 (drive).
5. Direct labels; running gear sits between the side-plate pairs.

FLAIL MOWER 500 / UC-V2

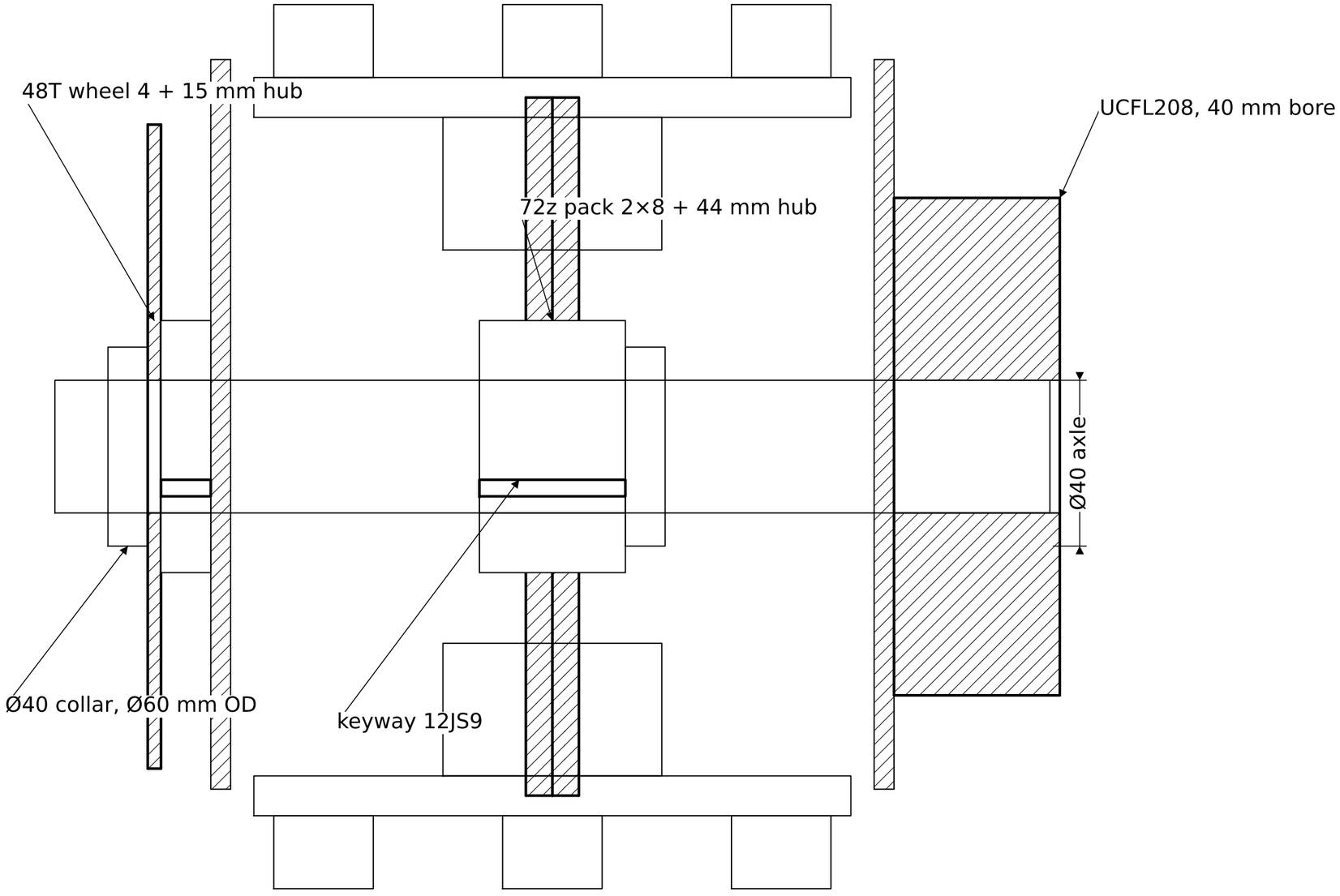
TOP VIEW - BOTH SIDE PLATES

DWG No.
UCV2-GA-005
REV A

MATERIAL	QTY	UNITS: mm
-	1	1st ANGLE

SCALE	SIZE	SHEET	DATE	DRAWN
1:6	A3	4 / 15	2026-10-29	OC

SECTION B-B (1:1.5)

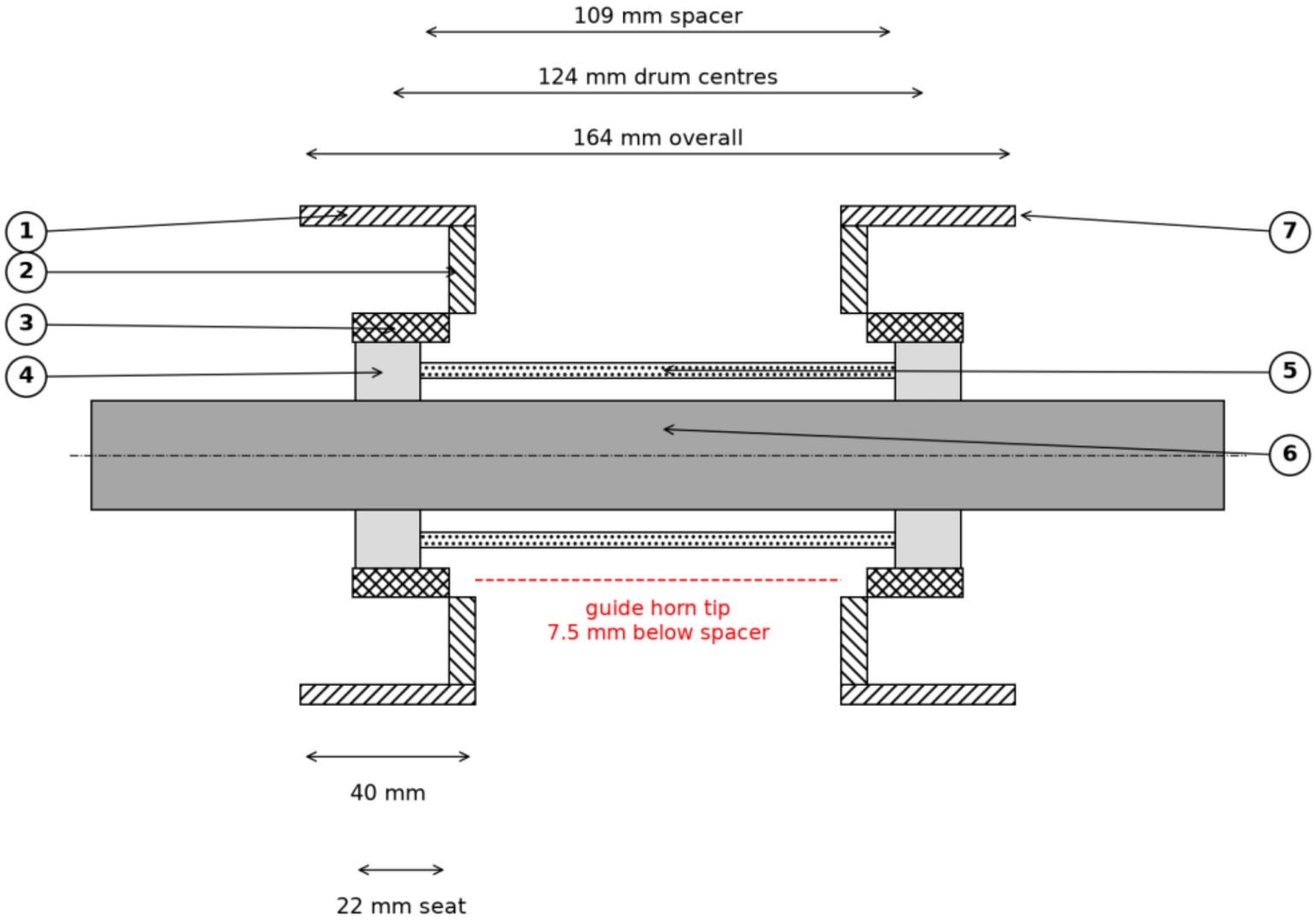


NOTES

1. Section at x=-500.6, looking forward. Local y=0 at track centre.
2. 48T y=-120 (global 140); 72z y=0 (global 260); axle z=133.3.
3. Axle Ø40 L300: keyways y=142..157 and 238..282.
4. UCFL208 363..413; axle end 410 -> 47 mm engagement.
5. Ø40 collars (OD60) trap the 48T wheel and the 72z hub.
6. Chain plane y=140 (global), 120 inboard of track centre.

FLAIL MOWER 500 / UC-V2 **SECTION B-B - DRIVE STATION**			DWG No. **UCV2-GA-004** REV A
MATERIAL EN8 / IS2062	QTY 1 station		**UNITS: mm** 1st ANGLE
SCALE 1:1.5	SIZE A3	SHEET 3 / 15	DATE 2026-10-29 / DRAWN OC

ROLLER Ø114.3 × 40 mm — one station, two identical drums



ROLLER

1 PIPE Ø114.3 × 4.5 wall × 40 mm

2 BACKPLATE 6 mm, welded inside pipe

3 BEARING SEAT 65/52 × 22 mm, welded to backplate, 52H7

4 6205-2RS (25 × 52 × 15 mm), press fit

5 SPACER 42.4/35.5 × 109 mm, loose, between bearing inner races

6 SHAFT Ø25 × 260 mm, press fit

7 OPEN OUTER END (cap optional)

Two identical drums at ±62 mm.

NOTES

1. V3 simple spool: 2 identical drums per station at ±62 mm.
2. Pipe + backplate + welded 65/52 × 22 mm seat; 52 H7 after welding.
3. 6205 outer race press in seat; inner race press on Ø25 shaft.
4. Loose 42.4/35.5 × 109 mm spacer between bearing inner races.
5. Central weld omitted; outer end open.
6. Bearing at tread centre; guide horn clearance 7.5 mm.

FLAIL MOWER 500 / UC-V2

ROLLER STATION V3 - SIMPLE SPOOL

DWG No. **UCV2-P-002** REV A

MATERIAL: MS / E250 / EN8

QTY: 12 drums

UNITS: mm

1st ANGLE

SCALE: NTS

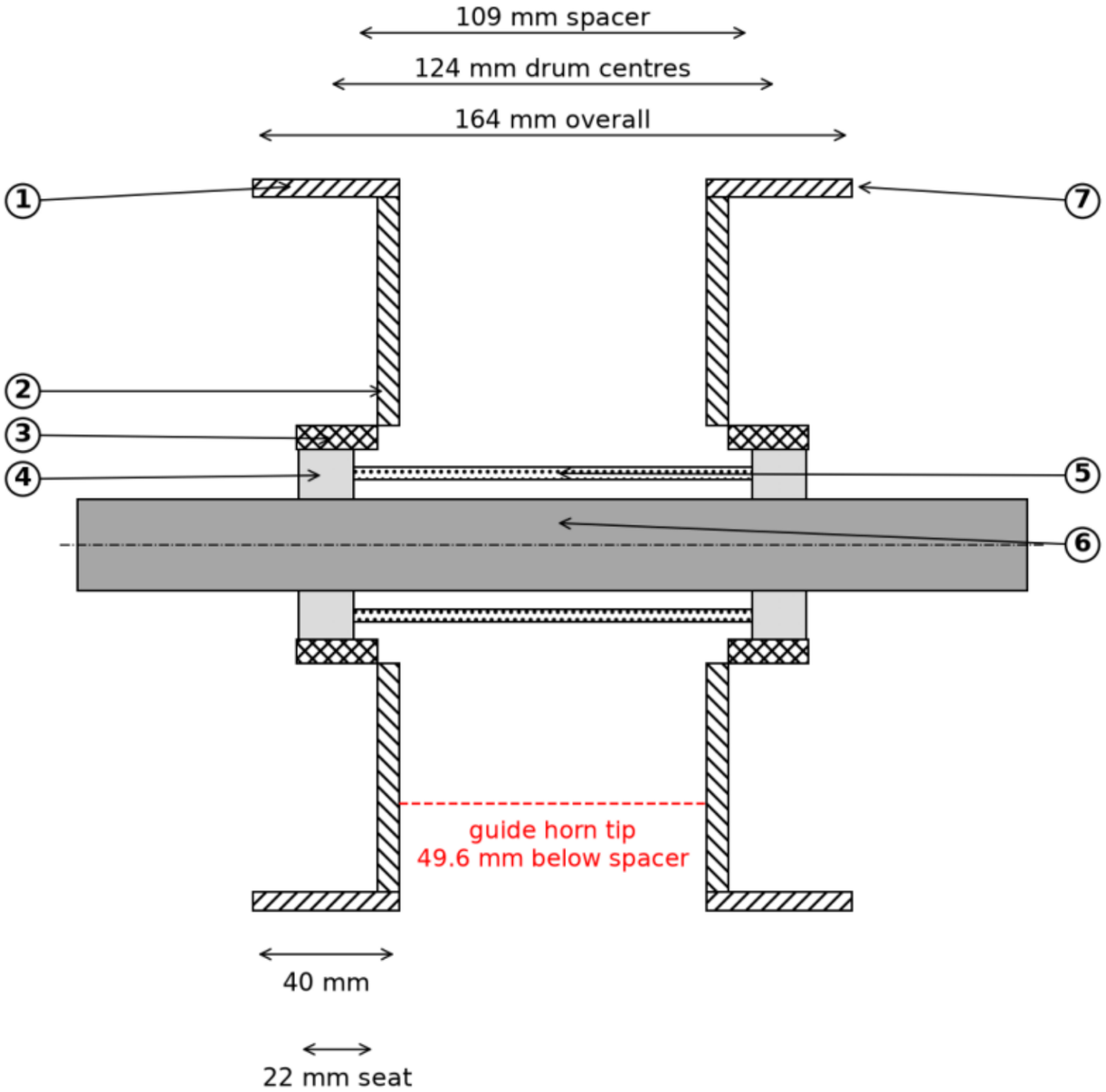
SIZE: A3

SHEET: 6 / 15

DATE: 2026-10-29

DRAWN: OC

IDLER Ø200 × 40 mm — one station, two identical drums



IDLER

1 PIPE Ø200.0 × 5.0 wall × 40 mm

2 BACKPLATE 6 mm, welded inside pipe

3 BEARING SEAT 65/52 × 22 mm, welded to backplate, 52H7

4 6205-2RS (25 × 52 × 15 mm), press fit

5 SPACER 42.4/35.5 × 109 mm, loose, between bearing inner races

6 SHAFT Ø25 × 260 mm, press fit

7 OPEN OUTER END (cap optional)

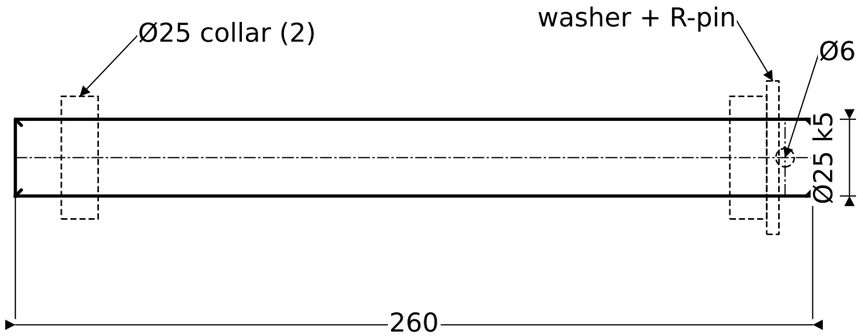
Two identical drums at ±62 mm.

NOTES

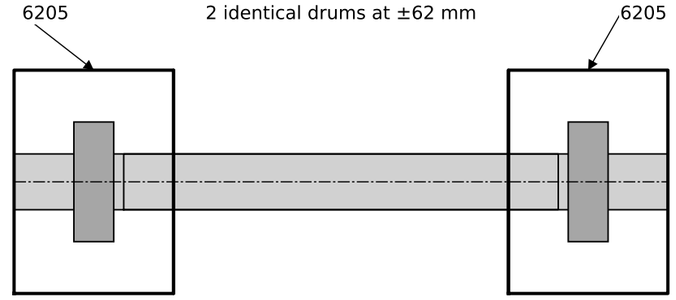
1. V3 simple spool: 2 identical drums per station at ±62 mm.
2. Ø200 × 5 mm pipe + backplate + welded 65/52 × 22 mm seat.
3. 6205 press fits; loose 42.4/35.5 × 109 mm spacer.
4. Central weld omitted; outer end open.
5. Idler pipe bore clears the guide horns by 24.2 mm.
6. Same Ø25 shaft/collars as the roller station.

FLAIL MOWER 500 / UC-V2 **IDLER STATION V3 - SIMPLE SPOOL**			DWG No. **UCV2-P-003** REV A
MATERIAL MS / E250 / EN8	QTY 4 drums		**UNITS: mm** 1st ANGLE
SCALE NTS	SIZE A3	SHEET 7 / 15	DATE 2026-10-29 / DRAWN OC

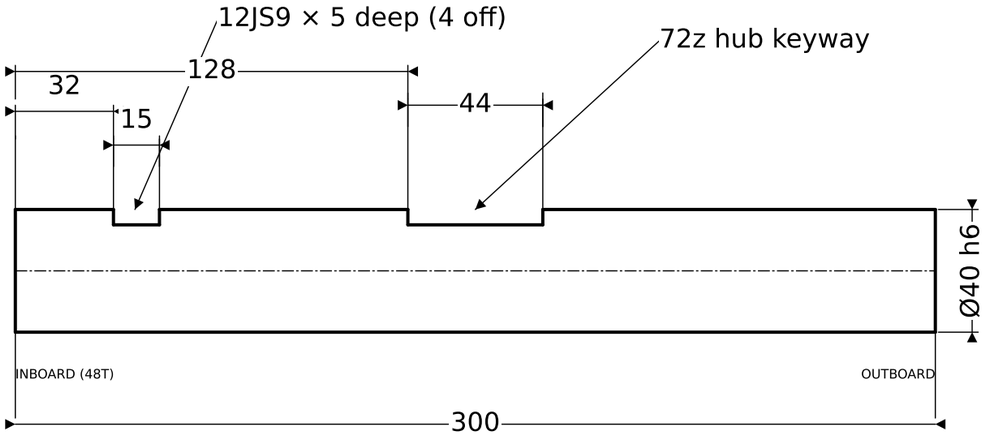
Ø25 SHAFT (1:2)



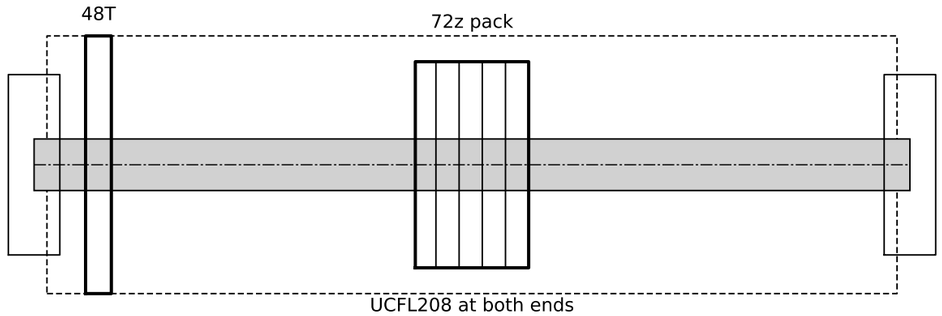
WHERE USED - ROLLER / IDLER SPOOL (context, NTS)



Ø40 AXLE (1:2)



WHERE USED - DRIVE STATION (context, NTS)

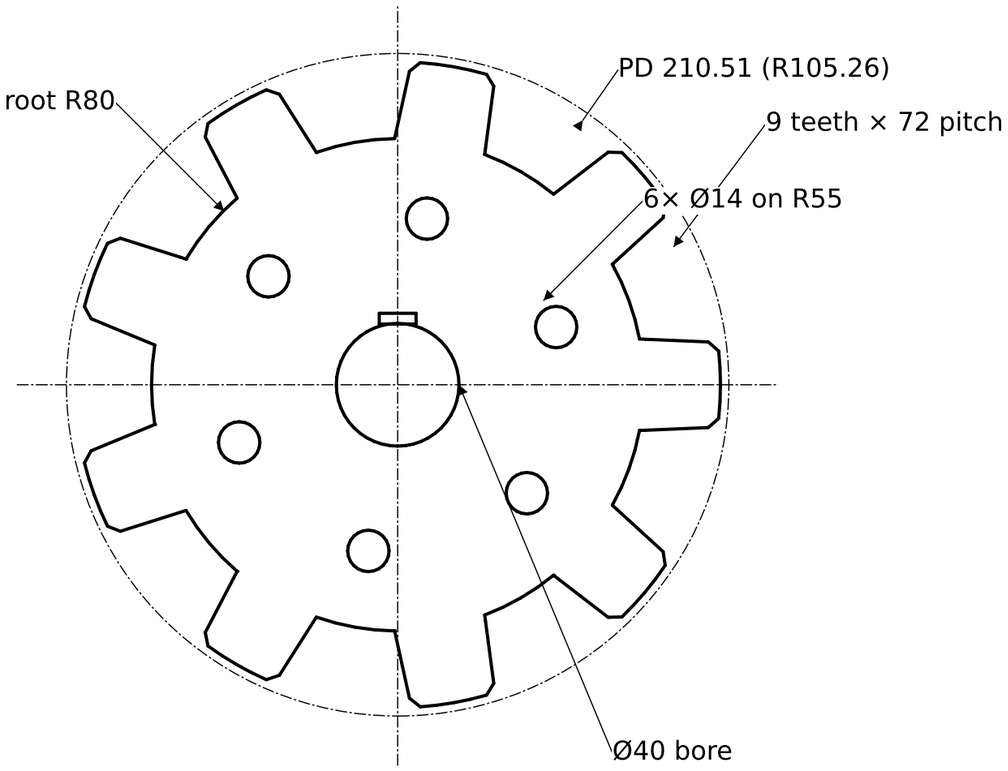


NOTES

1. Ø25: 8 off L260 mm, EN8; seats Ø25 k5, ends g6; Ø6 hole 9 mm O/B.
2. Ø40: 2 off L300 mm, EN8, h6; 4 keyways 12JS9 × 5 deep.
3. Keyways: 32..47 and 128..172 mm from 48T end.
4. Collars/washer/R-pin: COTS, position only.
5. Chamfer 1×45 both ends; protect keyways for paint.

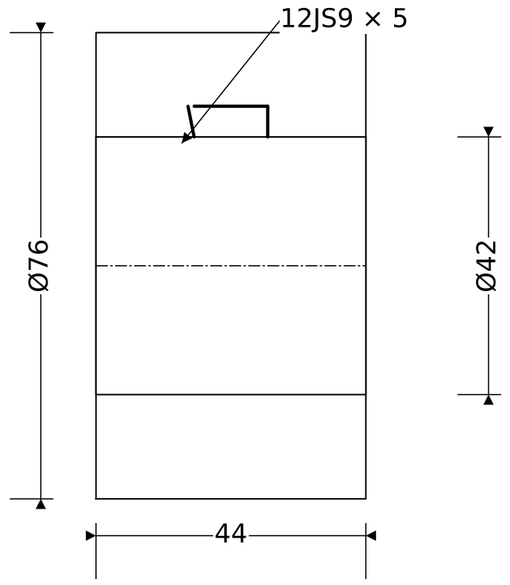
FLAIL MOWER 500 / UC-V2 **SHAFT Ø25 × 260 & AXLE Ø40 × 300**			DWG No. **UCV2-P-004** REV A
MATERIAL EN8 (bright)	QTY 8 + 2		**UNITS: mm** 1st ANGLE
SCALE 1:2	SIZE A3	SHEET 8 / 15	DATE 2026-10-29 / DRAWN OC

SPROCKET 72z9 (1:2)



2 plates/side pack, 8 mm each; 6× Ø14 mount holes

HUB (1:1)



6 rings × 8 mm, faced

DRIVE PARTS

part	description	mat	qty
72z9	9T × 72, PD210.51, Ø40, 8 mm	E250	4
hub76	Ø76/Ø42 × 44, 12JS9	E250	2

NOTES

1. 72 mm pitch, 9 teeth, PD 210.51; laser 8 mm E250, 2 plates/side pack.
2. Tooth profile per sprocket_72z9.dxf; tip R105.26, root R80.
3. Bore Ø40; hub carries the 12JS9 keyway (outsource).
4. Hub: 6 rings Ø76/Ø42 × 8 mm, stacked 48, faced 44.
5. Pack 2 × 8 = 16 mm wide, centred on track (y=±260); hub 238..282 mm.

FLAIL MOWER 500 / UC-V2

SPROCKET 72z9 & HUB 76/42

DWG No. **UCV2-P-005** REV A

MATERIAL: E250 8 mm

QTY: 4 + 2

UNITS: mm

1st ANGLE

SCALE: 1:2 / 1:1

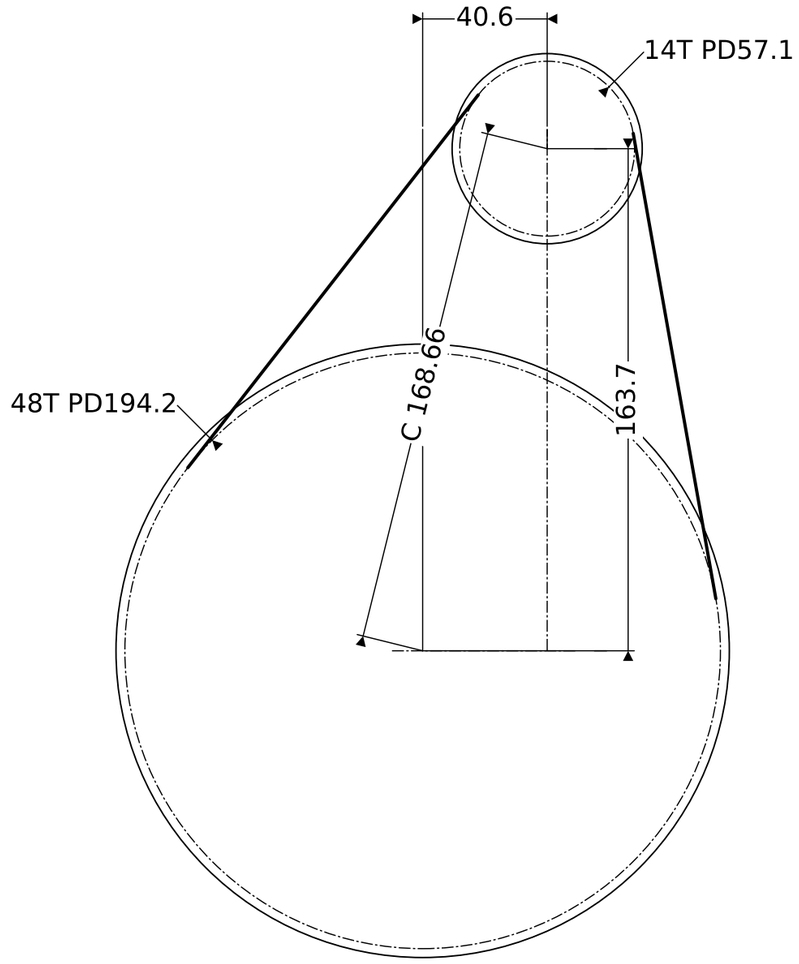
SIZE: A3

SHEET: 9 / 15

DATE: 2026-10-29

DRAWN: OC

CHAIN GEOMETRY (1:2)



CHAIN DATA

item	value	item	value
pitch	12.7 mm (428)	links	60
14T PD	57.07 mm	48T PD	194.20 mm
ratio	3.43	pull	1670 N
C	168.66 mm	sag	10-15 mm

NOTES

1. 428 pitch 12.7 mm; 14T pinion → 48T wheel, ratio 3.43.
2. 60 links at C=168.66 mm (rail slots take up 10-15 mm sag).
3. Chain plane y=±140; pinion and 48T wheel both on 140.
4. Static pull 835 N at 81 Nm; design pull 1670 N; SF 12 on a 20 kN chain.

FLAIL MOWER 500 / UC-V2

CHAIN DRIVE 14T/48T

DWG No.
UCV2-P-006
REV A

MATERIAL
428 chain

QTY
2 sets

UNITS: mm
1st ANGLE

SCALE
1:2

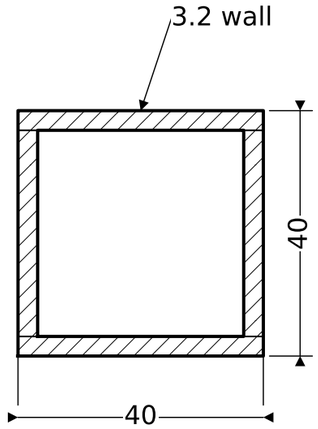
SIZE
A3

SHEET
10 / 15

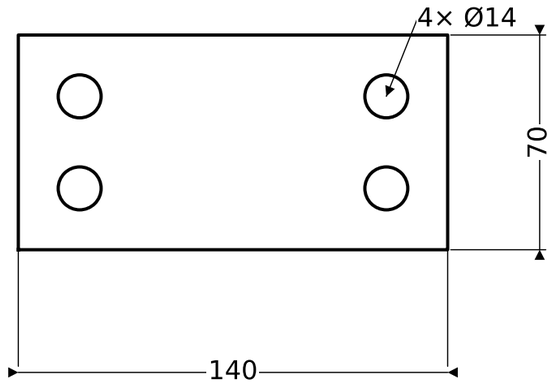
DATE
2026-10-29

DRAWN
OC

SHS40 (1:1)

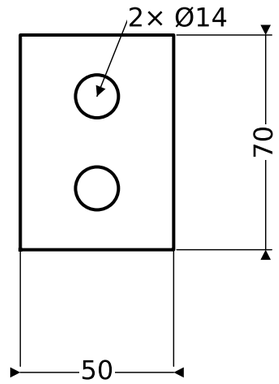


FLANGE MID (1:2)



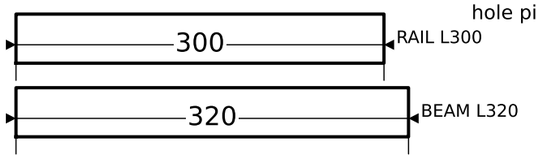
hole pitch 100 × 30

FLANGE FRONT (1:2)



hole pitch 30

BEAM L320 / RAIL L300 (1:5)



BOLT SCHEDULE

joint	flange	qty	M12
mid	140 × 70 × 8	4	16
rear	140 × 70 × 8	2	8
front	50 × 70 × 8	2	4
total	-	8	28

NOTES

1. All beams SHS40 × 40 × 3.2; 4× L320 fitted; 1 spare in stock.
2. Rails 4× L300 at z210..250; motors sit at z250.
3. Flanges: 4 mid + 2 rear 140 × 70 × 8 (4×M12); 2 front 50 × 70 × 8 (2×M12).
4. 28× M12 total (14/side): mid 16, rear 8, front 4.
5. TVS 8.8, 80 Nm, nylock + hardened washer + paint mark.
6. Weld 3 mm fillet tube-to-flange, stitch then continuous.

FLAIL MOWER 500 / UC-V2

FRAME SECTIONS, FLANGES & BOLT SCHEDULE

DWG No. **UCV2-P-007**

REV A

MATERIAL: SHS40 × 3.2 / E250 8 mm

QTY: as listed

UNITS: mm

1st ANGLE

SCALE: 1:1 / 1:2

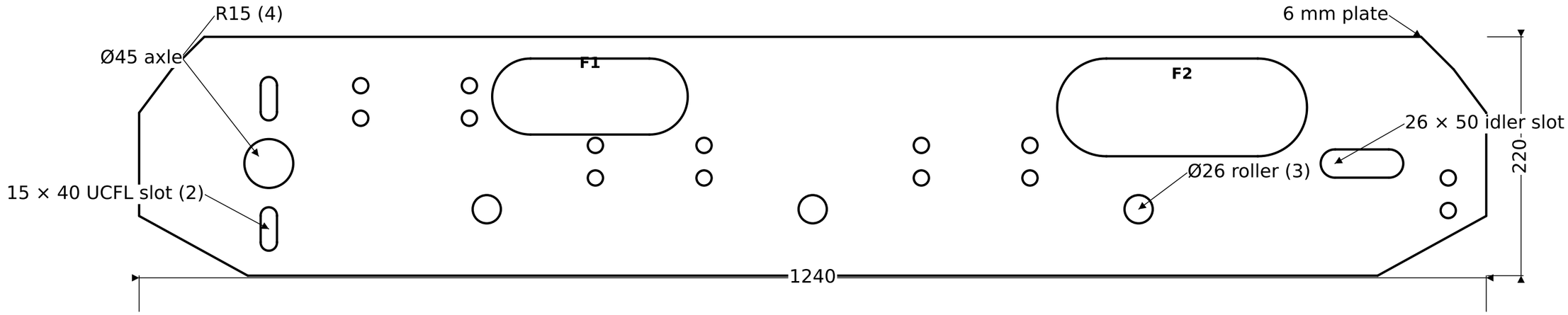
SIZE: A3

SHEET: 11 / 15

DATE: 2026-10-29

DRAWN: OC

FLAT PATTERN (1:4)



DATUM: X0 = MID, Z0 = PLATE BOTTOM (z = local y + 30)

HOLE TABLE (mm)

ID	x	z	size	qty
A	-300 / 0 / +300	91.2	Ø26	3
B	+480.6	133.3	26 × 50	1
C	-500.6	133.3	Ø45	1
D	-500.6	73 / 193	15 × 40	2
E	per P-007	120..205	Ø14	14
F1	-280..-130	160..230	150 × 70	1
F2	250..430	140..230	180 × 90	1

NOTES

1. 4 off: inner + outer per side; one DXF, mirror L/R.
2. Datum: X=0 mid, Z=0 plate bottom; table z global.
3. Break edges 1×45; corner R15 laser.
4. Seats: Ø26 roller (3), Ø45 axle, 26 × 50 idler slot, 15 × 40 UCFL slots.
5. 14 × Ø14 beam-flange holes (P-007).
6. 2 obround pockets 20R; no ribs (6 mm dent-governed).

FLAIL MOWER 500 / UC-V2

SIDE PLATE - FLAT PATTERN

DWG No. **UCV2-P-001**

REV A

MATERIAL: IS2062 E250Br 6 mm

QTY: 4 (2 in+2 out)

UNITS: mm

1st ANGLE

SCALE: 1:4

SIZE: A3

SHEET: 5 / 15

DATE: 2026-10-29

DRAWN: OC

PARTS LIST (as drawn)

#	cat	part	sheet	qty
1	structure	side plate 1240 × 220 × 6 mm	P-001	4
2	running	roller drum v3 (pipe+backplate+seat)	P-002	12
3	running	idler drum v3 (pipe+backplate+seat)	P-003	4
4	running	spacer 42.4/35.5 × 109 mm	P-002/3	8
5	running	shaft Ø25 × 260 mm	P-004	8
6	running	collar Ø25	P-009	16
7	running	washer Ø25/OD50	P-009	8
8	running	pin Ø6 × 40 mm	P-009	8
9	drive	axle Ø40 × 300 mm	P-004	2
10	drive	collar Ø40 OD60	P-009	4
11	drive	sprocket 72z9, 8 mm	P-005	4
12	drive	hub 76/42 stack	P-005	2
13	structure	beam SHS40 L320	P-007	4
14	structure	rail SHS40 L300	P-007	4
15	structure	flange mid 140 × 70 × 8	P-007	6
16	structure	flange front 50 × 70 × 8	P-007	2
17	structure	deck pad 80 × 80 × 20	P-008	4
18	structure	lift lug 10 × 60 × 80	P-008	2
19	drive	tension lug 30 × 24 × 40	P-009	4
20	drive	tension rod M12 × 110	P-009	4
21	drive	rod eye 70 × 24 × 50	P-009	4
22	structure	spacer tube Ø48 × 3.2 × 200	P-007	4
23	running	6205-2RS	P-002/3	16
24	drive	48T chainwheel (adapted)	P-006	2
25	drive	UCFL208	GA-004	4
26	drive	motor 1.2 kW + gearbox	GA-001	2
27	drive	428 chain (60L)	P-006	2
28	track	track 180 × 72 × 37	GA-001	2
29	misc	M12 nut (tension)	P-009	8
30	misc	fasteners/paint/misc	-	1

NOTES

1. Quantities are per machine (2 tracks).
2. Laser parts: 7 DXFs (plate, sprocket, 2 backplates, hub76, 2 flanges).
3. Outsource: 8 keyways + 2× 48T bore adapt; laser + machine shop only.
4. TVS 8.8 fasteners; 80 Nm structural M12; nylock + paint mark.
5. Sheet numbers refer to the drawing where the part is detailed.

FLAIL MOWER 500 / UC-V2

PARTS LIST / BOM

DWG No. **UCV2-P-010** REV A

MATERIAL: as listed

QTY: 1 machine

UNITS: mm

1st ANGLE

SCALE: -

SIZE: A3

SHEET: 14 / 15

DATE: 2026-10-29

DRAWN: OC